

Analyse and report on the distribution of data from primary and secondary sources using random and non-random sampling techniques to select and study samples

Learning intention

- analyse and report on the distribution of data from primary and secondary sources using random and non-random sampling techniques to select and study samples.

Teach and model

- investigating different methods of sampling used to collect data, considering the source and size of samples.
- defining and distinguishing between probabilistic terms such as random, sample space, sample and sample distribution.
- comparing the sampling methods of simple random, systematic, stratified, quota, clustered or convenience, or judgement.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.